MTEC 3501 — From Issues to Execution Systems

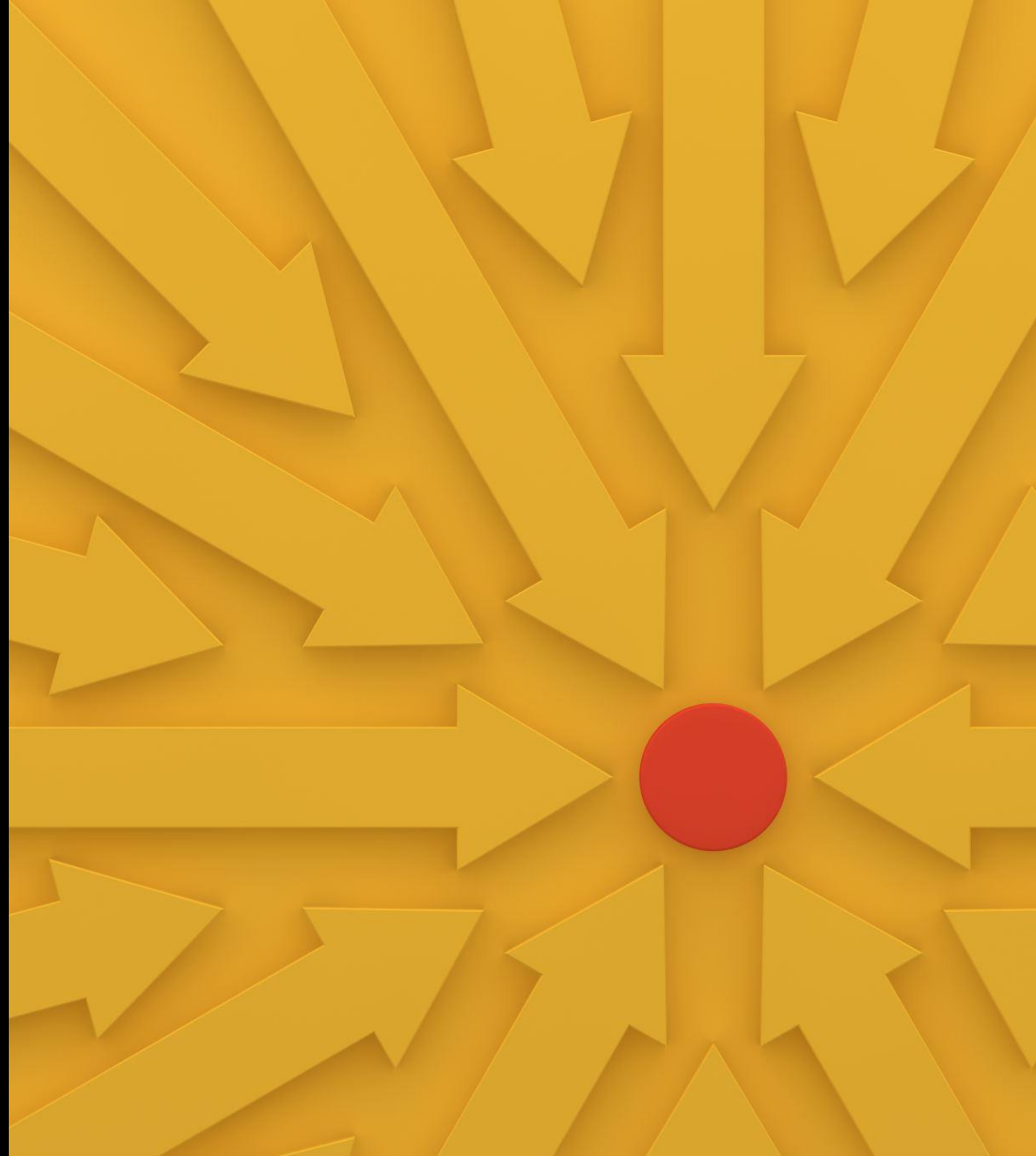
Milestones, Dependencies, and Project Boards

Where We Are

- Speculate → Research → Design → Make
- You have Issues (transformations)
- Now we add time + state

Issues as Transformations

- Source → Vector → Destination
- Issues are not tasks



What an Issue actually is

Current State (problem)

Transformation (what you do)

Defined Outcome (testable result)

Task vs Transformation

Task:

"Work on audio system"

Issue (Transformation):

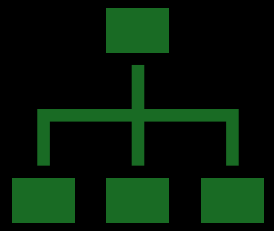
Source: No signal flow defined

Vector: Design + implement signal path

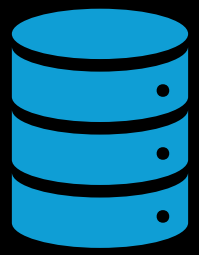
Destination: Input → output pipeline working and documented

What’s Missing

Milestones (time)



Board (state)



Dependencies (causality)

System Model

- Issues + Milestones + Board + Dependencies = Execution System

What is a Milestone?

- Time-bound container
- Defines target system state

Milestones vs Issues

- Issue = what
- Milestone = when

Building Milestones into your project

Final Milestone First:

Start with the End

We do not begin by asking:

- What should we do next?

We begin by asking

- What must exist at the end?

Final Milestone: Presentation-Ready System

Core Statement

At the end of the semester, your project must exist as a **complete, communicable system**

Presentation Structure:

System (PoC)

- Functional within scope
- Stable enough to demonstrate
- Core interactions work

Communication (Presentation)

- Slides complete
- Narrative clear
- Demo integrated

Documentation (Repository)

- README explains project
- Artifacts included
- Work traceable through issues

Example Milestone

- Prototype v0:
- - Signal flow
- - Working test
- - Basic interaction

Common Mistake

- Bad: Work on project
- Good: Defined functional outcome

Dependencies

- Define order of work
- Reveal constraints

Dependency Example

- Design → Implementation
- If design fails, implementation fails

Project Board

- Real-time system state model
- Not a checklist

Board States

- Backlog → Ready → In Progress → Blocked → Review → Done

GitHub Structure

- Issues, Milestones, Projects, Links

Create Milestone

- Issues → Milestones → New
- Add title, date, description

Assign Issues

- Open issue → assign milestone

Dependencies in GitHub

- Use: Depends on # / Blocks #

Create Project Board

- Projects → New → Board

Add Columns

- Backlog, Ready, In Progress, Blocked, Review, Done

Add Issues to Board

- Add items → select issues

Workflow Rules

- Define Ready and Done conditions

Views

- Table, Board, Filters

Example Views

- By milestone, status, blocked, component

System Check

- Can you see what’s next, blocked, delayed?

In-Class Task

- Create milestones
- Assign issues
- Add dependencies
- Build board

Final Thought

- A project is the system that produces your idea